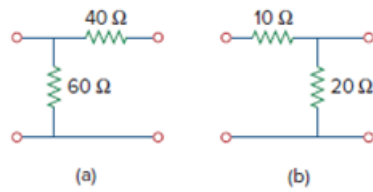


Problem

Find the h parameters for the networks in Fig. 19.88.

Figure 19.88



Step-by-step solution

Step 1 of 17

Refer to Figure 19.88(a) in the textbook.

Re-draw the circuit diagram:

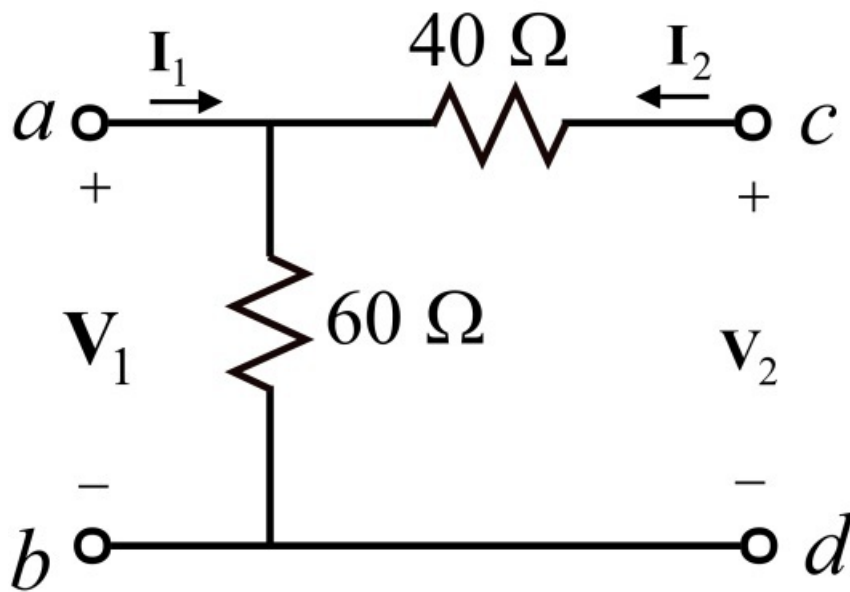


Figure 1

Step 2 of 17

Write the h -parameters of the network.

$$V_1 = h_{11}I_1 + h_{12}V_2$$

$$I_2 = h_{21}I_1 + h_{22}V_2$$

The h -parameters of the network are:

$$h_{11} = \left. \frac{V_1}{I_1} \right|_{V_2=0}$$

$$h_{12} = \left. \frac{V_1}{V_2} \right|_{I_1=0}$$

$$h_{21} = \left. \frac{I_2}{I_1} \right|_{V_2=0}$$

$$h_{22} = \left. \frac{I_2}{V_2} \right|_{I_1=0}$$

Step 3 of 17

Determine the parameter, h_{11} .

From Figure 1, short-circuit the terminals c and d .

Re-draw the circuit diagram.

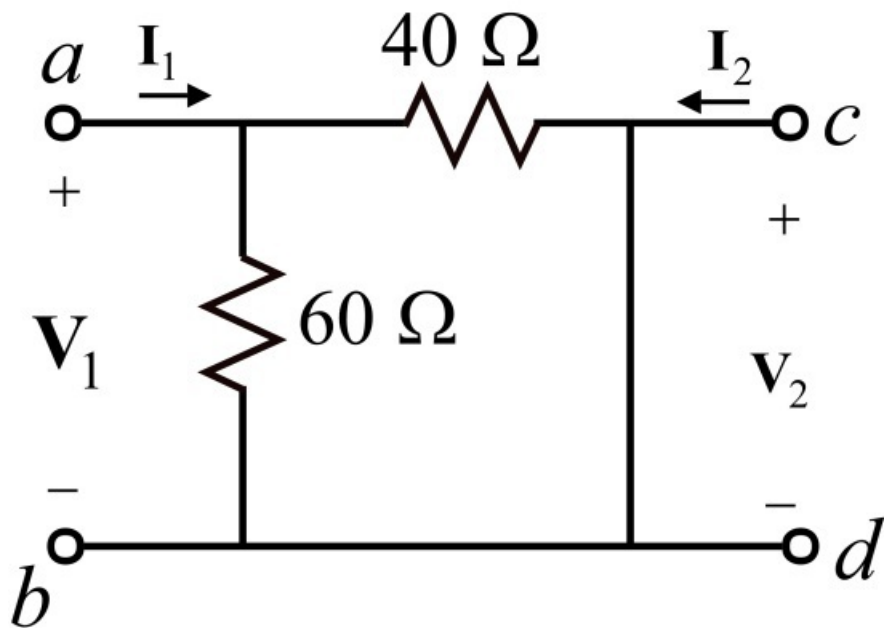


Figure 2

Step 4 of 17

From Figure 2, the resistance $60\ \Omega$ and $40\ \Omega$ are in parallel. Determine the equivalent resistance.

$$\begin{aligned} R_{eq} &= 60 \parallel 40 \\ &= \frac{(60)(40)}{60 + 40} \\ &= 24 \end{aligned}$$

The value of $\frac{V_1}{I_1}$ is,

$$\begin{aligned} \frac{V_1}{I_1} &= R_{eq} \\ &= 24\ \Omega \end{aligned}$$

The value of h_{11} is,

$$\begin{aligned} h_{11} &= \frac{V_1}{I_1} \\ &= 24\ \Omega \end{aligned}$$

Therefore, the value of h_{11} is $\boxed{24\ \Omega}$.

Step 5 of 17

Determine the parameter, h_{21} .

From Figure 2, apply current division rule and determine the current through $40\ \Omega$ resistance.

$$\begin{aligned} I_{40\ \Omega} &= I_1 \left(\frac{60}{60 + 40} \right) \\ &= I_1 \left(\frac{60}{100} \right) \\ &= 0.6I_1 \end{aligned}$$

Step 6 of 17

From Figure 2, $I_{40\ \Omega} = I_2$.

Therefore,

$$\begin{aligned} I_2 &= 0.6I_1 \\ \frac{I_2}{I_1} &= 0.6 \end{aligned}$$

Therefore, the value of h_{21} is $\boxed{0.6}$.

Step 7 of 17

Determine the parameter, h_{12} .

From Figure 1, open-circuit the terminals a and b .

Re-draw the circuit diagram.

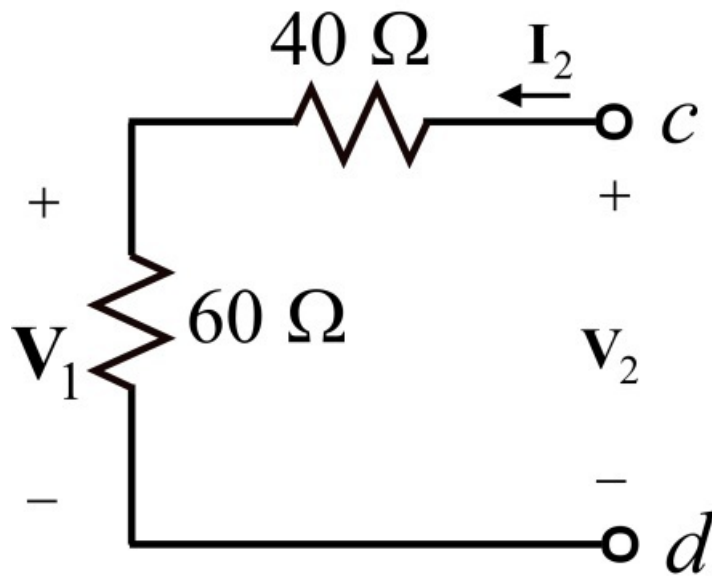


Figure 3

Step 8 of 17

From Figure 3, apply voltage division rule.

$$\begin{aligned} V_1 &= V_2 \left(\frac{60}{60 + 40} \right) \\ &= V_2 \left(\frac{60}{100} \right) \\ &= 0.6V_2 \end{aligned}$$

$$\frac{V_1}{V_2} = 0.6$$

The value of h_{12} is,

$$\begin{aligned} h_{12} &= \frac{V_1}{V_2} \\ &= 0.6 \end{aligned}$$

Therefore, the value of h_{12} is .

Step 9 of 17

Determine the parameter, h_{22} .

From Figure 3, apply ohms law.

$$\begin{aligned}V_2 &= I_2 (40 + 60) \\ &= I_2 (100)\end{aligned}$$

$$\begin{aligned}\frac{I_2}{V_2} &= \frac{1}{100} \\ &= 0.01 \text{ S}\end{aligned}$$

Therefore, the value of h_{22} is 0.01 S .

The h -parameters of the network are:

$$h = \begin{bmatrix} 24 \Omega & 0.6 \\ 0.6 & 0.01 \text{ S} \end{bmatrix}$$

Step 10 of 17

Refer to Figure 19.88(b) in the textbook.

Re-draw the circuit diagram:

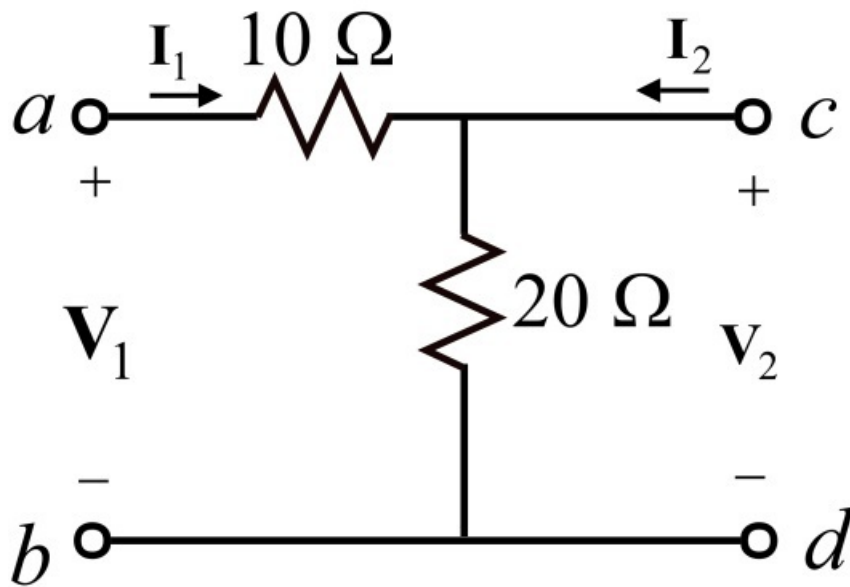


Figure 4

Step 11 of 17

Determine the parameter, h_{11} .

From Figure 4, short-circuit the terminals c and d .

Re-draw the circuit diagram.

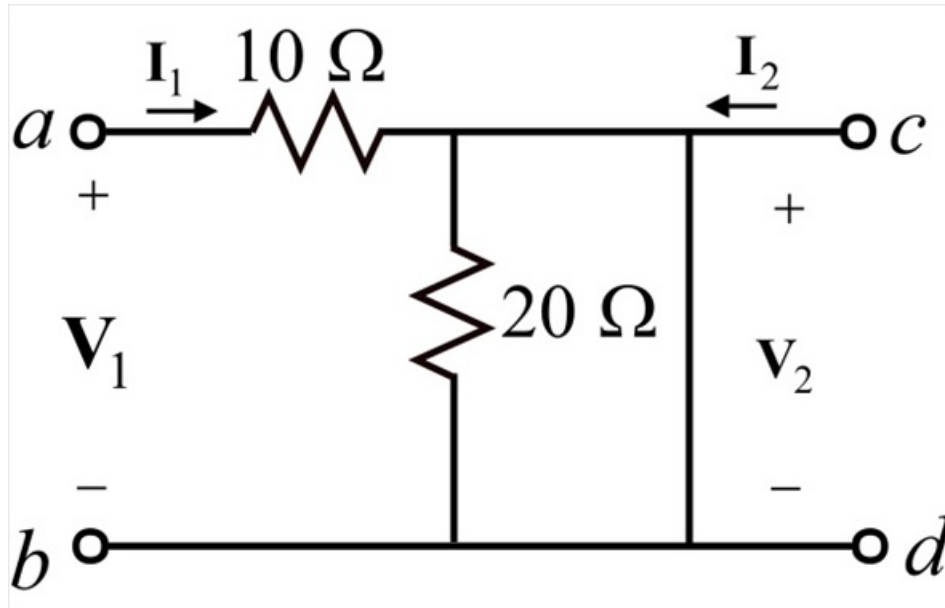


Figure 5

Step 12 of 17

From Figure 5, the $20\ \Omega$ resistance is short circuited.

Re-draw the circuit diagram.

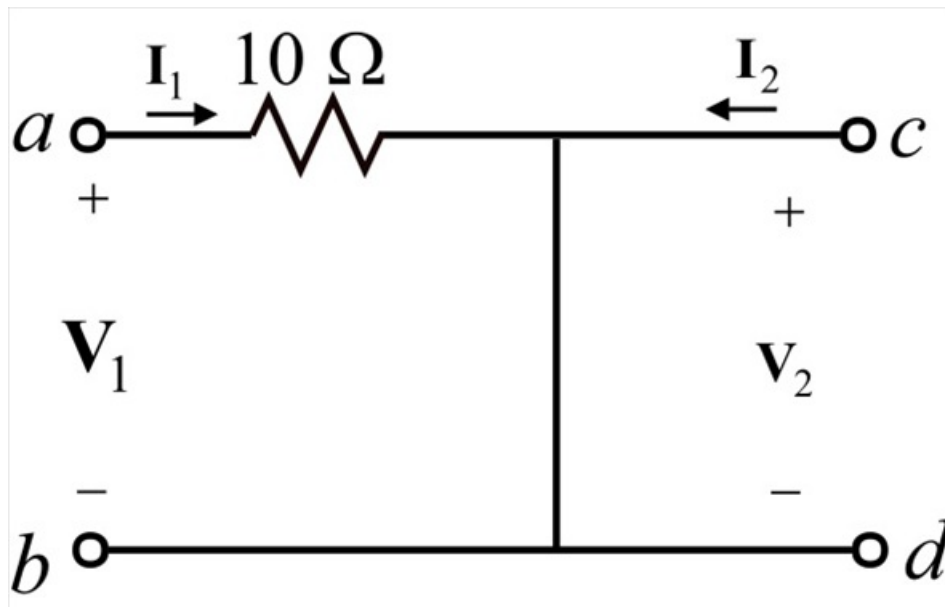


Figure 6

Step 13 of 17

From Figure 6, determine the value of $\frac{V_1}{I_1}$.

$$\frac{V_1}{I_1} = 10 \, \Omega$$

The value of h_{11} is,

$$\begin{aligned} h_{11} &= \frac{V_1}{I_1} \\ &= 10 \, \Omega \end{aligned}$$

Therefore, the value of h_{11} is $\boxed{10 \, \Omega}$.

Step 14 of 17

Determine the parameter, h_{21} .

From Figure 6, $I_1 = I_2$

Therefore,

$$\frac{I_2}{I_1} = 1$$

Therefore, the value of h_{21} is $\boxed{1}$.

Step 15 of 17

Determine the parameter, h_{12} .

From Figure 4, open-circuit the terminals a and b .

Re-draw the circuit diagram.

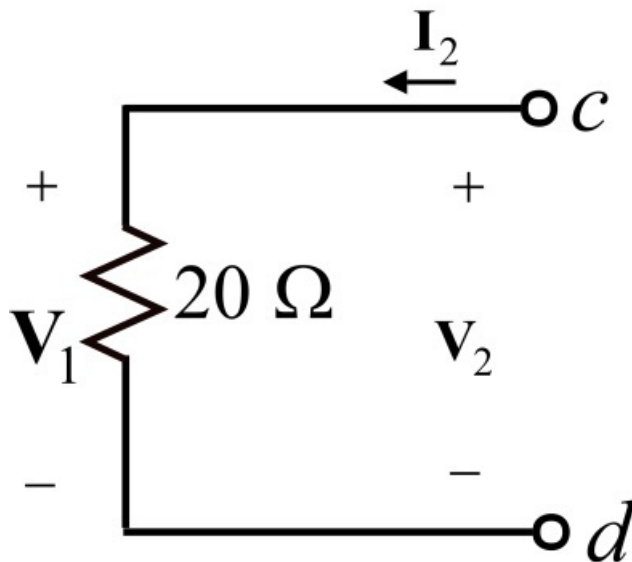


Figure 7

Step 16 of 17

From Figure 7, $V_1 = V_2$

The value of h_{12} is,

$$h_{12} = \frac{V_1}{V_2}$$
$$= 1$$

Therefore, the value of h_{12} is .

Step 17 of 17

Determine the parameter, h_{22} .

From Figure 7, determine the value of $\frac{I_2}{V_2}$.

$$\frac{I_2}{V_2} = \frac{1}{20}$$
$$= 0.05 \text{ S}$$

Therefore, the value of h_{22} is .

The h -parameters of the network are:

$$h = \begin{bmatrix} 10 \, \Omega & 1 \\ 1 & 0.05 \, \text{S} \end{bmatrix}$$